

Retained complex totals with compact controls

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Abstract

A complex producer combines disjoint rectangles with retained pair roots and doubled point totals, then scatters them at one common low/full frame. At $h = 24$ this supports $1 - \sigma = 750/10^{11}$. With the existing compact-control construction and paired-bit producer, the exact conditional witness is $\kappa = 591/10^{12} > 2^{-31}$, a factor $591/83 \simeq 7.12$ above the preceding compact-control witness. The original fixed-tape and analytic interfaces remain assumptions; this is not a full formal verification or runtime result.

The source is OpenAI’s *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

We retain Douglas Colkitt’s compact-control movement, layout and generalized guard arguments, paired-bit producer, Gaussian setup, and the original resampling, synthetic-transform, error and rounding interfaces.

0.1 A retained-total complex motif at ground size twenty-four

Use $h = 24$, $v = \binom{h}{3} = 2024$, $N = v^3$, and $m = h^3 = 13824$, independently of the retained paired-bit arity 125000. The scalar domain is $\mathbb{Z}[i, 1/2]$ and the binary label space is $F = \mathbb{F}_2^h$ with its ordinary dot product. Triple indicators t_T have norm one. The full space is $\mathcal{F} = F^{\otimes 3}$. All networks, labels and bases below are fixed finite data.

Side correction by shared source trees. The central scalar coefficient is $(|S \cap T| - 1)/2$. Its side correction is $+1/2$ when S, T are disjoint and $-1/2$ when $|S \cap T| = 2$, and zero otherwise. Central plus side is the identity. For the disjoint branch, recursively partition ordered disjoint triple pairs into rectangles. On $D(a, b)$, use a row-star partition, a column-star partition, or split the ground set and partition each source/target cardinality case by the product of its two smaller disjointness partitions. Each case is disjoint, and the selected recursive construction is an exact partition, not a cover. For a rectangle with a source triples and b targets, aggregate its source values by a balanced tree and retain one output use per target. It costs $a + b - 1$ roles. A singleton source retains its triple-line label. For a larger source family use coordinate labels of its source unions. If its full source union is the complete complement of any target, split that rectangle into singleton-edge rectangles to avoid an alternating output residual. Every repaired rectangle in the selected $h = 24$ plan has one target, so this repair preserves its role count. The repaired disjoint branch costs 329811 roles, with 123855 additions and 205956 outputs. Separately retain the lexicographic

*Contribution prepared with substantial OpenAI Codex assistance, including Sol 6.1 and Astra model audits. Original constructions are due to OpenAI and Douglas Colkitt. This is not independent human review.

source tree, grouped by least and second point, only for its global total below. For each fixed pair P , build a balanced tree on $x_{P \cup \{k\}}$, $k \notin P$. For target $P \cup \{r\}$ select maximal subtrees avoiding r , with coefficient $-1/2$. They partition the intersection-two neighbors, each with its unique shared pair. Their proper variable sets leave a spare point beyond r .

Retained roots and doubled totals. Retain every full pair root and the global disjoint-tree root:

$$A_{ij} = \sum_{k \notin \{i,j\}} x_{\{i,j,k\}}, \quad C_* = \sum_T x_T.$$

For each $i < h-1$, use a balanced tree on its $h-1$ incident pair roots to retain

$$D_i = \sum_{j \neq i} A_{ij} = 2 \sum_{T \ni i} x_T.$$

The factor two counts the two other points of each triple. These point-tree supports overlap; they are scalar sums with their actual integer multiplicities. Scatter the h retained totals by

$$(Rz)_S = \begin{cases} \frac{1}{4} \sum_{i \in S} D_i - \frac{1}{2} C_*, & h-1 \notin S, \\ C_* - \frac{1}{4} \sum_{i < h-1, i \notin S} D_i, & h-1 \in S. \end{cases}$$

On source contributions, the omitted point total equals $3C_* - \frac{1}{2} \sum_{i < h-1} D_i$. Substituting gives the central coefficient $(|S \cap T| - 1)/2$. No such relation is assumed for arbitrary initial scratch.

Binary labels and orthonormal residuals. For a nondegenerate binary subspace U , its characteristic vector is $P_U c$, where $c = (1, \dots, 1)$, since $x \cdot x = x \cdot c$. If $U \subset V$ are nondegenerate, the orthogonal difference $V \cap U^\perp$ is nonalternating precisely when the characteristic vector of V is outside U . Every nondegenerate nonalternating binary space admits an orthonormal basis: while more than one dimension remains, choose a unit vector different from the characteristic vector. Its complement remains nonalternating; the one-dimensional case terminates the induction.

Label each leaf by its triple line. A disjoint-tree addition with source union K has the coordinate label F_K . A triple-to-coordinate residual has characteristic $1_K + t_T \neq 0$; coordinate-to-coordinate residuals have coordinate unit bases. At a selected output $F_K \subset t_S^\perp$; the spare point gives a unit in the output residual. Selecting the whole complement of S would leave an alternating two-plane and is forbidden. Full coordinate nodes are retained for the global total; their retirement complement is zero and needs no unit witness.

A fixed-pair node with variable set K has the orthonormal basis $\{1_P + e_k : k \in K\}$. Its child residuals are the newly added basis vectors. Its selected output residual has a unit at the spare point outside P, K, r . Put $w_i = c + e_i$. Because h is even, $w_i \cdot w_j = [i = j]$. The full pair-root label is $U_{ij} = \text{span}\{w_k : k \neq i, j\}$: the sum of its original star basis vectors is $c + e_i + e_j$, converting each basis vector into w_k . Thus its complement has basis w_i, w_j . Label each addition forming D_i by

$$S_i = w_i^\perp = \text{span}\{w_k : k \neq i\}.$$

Two distinct incident pair-root labels span S_i , with unit-line residuals; later joins keep this label. Its characteristic is e_i and its complement is the unit line w_i . These witnesses cover activation, retirement and all new internal residuals, including nodes whose source union is full.

Exact reversible scalar wrapper. Use the outgoing-use/pivot compiler on the complete DAG, sharing identical source inputs and allocating distinct physical fanout roles. With C additions and q designated outputs it has $R_{\text{aux}} = C + q$ roles. Let L be its invertible mixer, V the source copy, and J the side injection. The coefficient identities give $(R + J)LV = I$. The chronological schedule

$$L, -R, -J, L^{-1}, V, L, R, J, L^{-1}, -V$$

subtracts $(R + J)Lz$, adds $(R + J)L(z + Vx)$, changes the target by x , and restores every arbitrary initial scratch value. Invert the whole schedule for the inverse shear. Forward, inverse with banks exchanged, forward gives $(X, Y) \mapsto (-Y, X)$ on every data pair, fixing every auxiliary role.

Common scatter frames in both orientations. At stage j write the original prefix decomposition $A = B \perp P$, where P and the future factor Q are norm-one tensor lines. Put

$$D_0 = (B \otimes F) \otimes Q, \quad D_1 = (A \otimes F) \otimes Q, \quad D_U = D_0 \perp (P \otimes U \otimes Q).$$

Early mixer and scatter gates use D_0 ; late cleanup uses D_1 . The forward source copy uses source-line frames, the middle mixer uses D_U , the entire retained-total scatter uses the common frame D_0 , and only then the side injection uses each target-complement frame. Consequently each D_i output descends once from D_{S_i} to D_0 , losing $h - 1$ dimensions; the global output descends from D_1 to D_0 , losing h . The scatter is an actual scalar gate touching all totals and data targets at one common frame; no decrease is free. The reverse scatter uses D_1 , followed by the complemented inverse mixer at $D_{U^\perp}$, giving the same orthogonal differences. The only local loss in either orientation is

$$\ell = (h - 1)^2 + h = 553.$$

The scalar target updates R, J commute, but their distinct frame assignments are essential. There is no direct edge $S_i \rightarrow t_S^\perp$. Tensoring local residual bases by the unit generators of P, Q preserves orthonormality. The additional data-boundary residuals are $B \otimes t_S^\perp \otimes Q$; coordinate units outside the earlier tensor support and the target triple give norm-one witnesses whenever nonzero. The original data-stage boundaries are unchanged.

Shared banks, endpoints and phase contract. Pair adjacent ground points and let $\pi(T)$ flip each point to its partner. This triple involution has intersection zero or two with its input. Match every complete stage-1 auxiliary bank (A, B) to the stage-3 bank $(B, \pi(A))$, keeping local physical role indices. Scalar restoration makes this valid for arbitrary shared scratch. The joining labels are

$$E = F \otimes U_A \otimes U_B, \quad H = (U_B \otimes U_{\pi(A)})^\perp \otimes F.$$

Here U_T is the triple line. The middle pairing vanishes, hence $E \subset H$. A tensor of coordinate units outside B in the first and third factors, and any coordinate unit in the second, lies in $H \cap E^\perp$ and has norm one. Its residual has dimension $m - 2h$. Replacing the two separate endpoint extensions of dimension $m - h$ by this join removes one role and exactly m kernel calls. Every surviving auxiliary still has source zero and sink full.

Data terminals remain $X : U_a \rightarrow \mathcal{F}$ and $Y : 0 \rightarrow U_a^\perp$, with u_a of weight 27. Orthogonal Hamming weight is additive modulo four, so each edge phase difference is one forward or inverse $aI + bX_v$ kernel per orthonormal residual basis vector. The original source/sink Z corrections, i^{27f} scalar, output sign removal and bank exchange give $C^{\otimes mf}$ on every role for every positive number f of columns. This is the pinned manuscript's binary phase argument applied to the residuals above, with no additional recursive child calls.

Exact counts and recurrence. The disjoint and intersection-two output counts are $q_0 = 205956$, $q_2 = 27600$. The rectangle count independently gives 123855 additions and 205956 output uses. Each fixed-pair tree on $n = h - 2$ variables has $q_P = n(n - 1) - \sum_{\text{internal}} (n - |K|)$ outputs. Retention activates all $v - 1$ global-tree additions, all $\binom{h}{2}(h - 3)$ pair additions and $(h - 1)(h - 2)$ point additions, or 8325 additions before rectangles. Thus $C = 132180$, $q = q_0 + q_2 + h = 233580$, and $R_{\text{aux}} = 365760$. The h totals are included in this count; no separate center bank is added.

$$W = 2N + 2v^2 R_{\text{aux}} = 3013310215168, \quad L_{\text{loss}} = 3v^2 \ell = 6796219584,$$

$$s = Wm - 2N + 2L_{\text{loss}} = 41655997423981952, \quad \eta = \frac{Wm - s}{Wm} = \frac{365}{5084246016}.$$

Proposition 1. *The retained-total complex motif has the all-role scalar and binary phase contracts above, with exactly s directional-kernel calls and recurrence saving $1 - \sigma = 750/10^{11}$.*

Indeed exact enclosure gives $\log m < 477/50$ and

$$\eta - \frac{750}{10^{11}} \frac{477}{50} = \frac{11935523}{49650840000000000} > 0;$$

$e^{-x} > 1 - x$ therefore gives $s/W < m^{1-750/10^{11}}$. There are $6696869422848 < 12W$ scalar gates, with coefficient magnitude at most one and denominator at most four. Saved-input evaluation costs at most two halvings and one addition per term, hence $3W^2$ operations per gate. Including inverse-child corrections,

$$36W^3 + 4s + 4W + 4 < 64(W + m + 1)^3 = E,$$

and $2 \leq s < m^5$. The retained generalized stopped-depth guard applies with $B = s + E$ and $C_1 = 5 - 4\beta + \zeta$.

Compact-control and assembly transfer. Instantiate the existing compact row reservations with this m, W, s . The totals are scalar roles; they introduce no address coordinates. Every recursive child still has exactly V/W volume and all complete compact fields. The written movement, repair, reservation and recurrence proofs give the same internal, leaf and reservation exponents

$$\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}, \quad \chi_{\text{leaf}} = \sigma + \beta(1 - \sigma), \quad \chi_{\text{reserve}} = \max\{1 - c, 0\}.$$

Keep $\tau = 1 - 296/10^{11}$, $\epsilon = 1999/10000$, $\beta = 1/1000$, $\zeta = 1/10000$, $\delta = 10^{-6}$ and $C_1 = 49961/10000$, and choose

$$c = 1, \quad \lambda = 1 - 2959/10^{12}, \quad \lambda' = 1 - 2958/10^{12}, \quad \kappa = 591/10^{12} > 2^{-31}.$$

All completed-layer inequalities and seven assembly margins are strict, with

$$\min_i g_i = g_3 = \frac{2956521}{5 \cdot 10^{15}}, \quad \min_i g_i - \kappa = \frac{1521}{5 \cdot 10^{15}} > 0.$$

Thus the retained assembly conditionally gives $T(n) = O(n(\log n)^{1-591/10^{12}})$, with saving 591/83 times the preceding $83/10^{12}$ witness. The fixed finite alphabet, fixed number of tapes, compact-control tape proofs and pinned analytic/multiplication interfaces remain premises. Exact finite checks do not prove the complete machine theorem.

Reproduction and verification boundary. Run `make verify` for the exact h24 coefficient map, retained integer multiplicities, both compiled frame directions, strict parameter comparisons, small dirty-scratch and phase-edge controls, and alternative patch checks. The certificate is `certificates/retained-complex-layer.json`; the patch is `patches/retained-complex-31.patch`, applied directly to the original pin. Build this note with `make retained-complex-note`. The complete fixed-tape and analytic multiplication theorem remains assumed.